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Q.4.43 The operation in the question is a sequence of modulation → DFT → conjugation → inverse DFT → modulation. The important result is that the final image is 180° rotated version of the original image.

Let the original image be $f(x,y)$ and assumes its size is $M \times N$

1. First multiplication by $(-1)^{x+y}$

$$g(x,y) = f(x,y)(-1)^{x+y}$$

The factor

$$(-1)^{x+y} = e^{j\pi(x+y)}$$

is a spatial modulation

By the DFT modulation shift property,

$G(U,V) = F((U-M/2)_M, (V-N/2)_N)$ for even M, N.

So, multiplication by $(-1)^{x+y}$ shifts the fourier transformal by half the image dimensions.

2. Compute the DFT

$$G(U,V) = F\{g(x,y)\}$$

3. Take the complex conjugate we obtain

$$G^*(U,V)$$

for a real image.

$$G^*(U,V) = G(-U,-V)$$

With indices impressed module M, N.

4. Compute the inverse DFT using the important DFT property

$$F^{-1}(G^*(U,V)) = g^*(-x,-y)$$

and since the image is real

$$g^*(x,y) = g(x,y)$$

therefore

$$F^{-1}\{G^*(U,V)\} = g(-x,-y)$$

Now substitute the definition of g

$$g(-x,-y) = F(-x,-y)(-1)^{-x-y}$$

Since

$$(-1)^{-x-y} = (-1)^{x+y}$$

We get

$$g(-x,-y) = F(-x,-y)(-1)^{x+y}$$

5. Multiply by $(-1)^{x+y}$ again

The final image is

$$h(x,y) = (-1)^{x+y} g(-x,-y)$$

Substituting

$$h(x,y) = (-1)^{x+y} F(-x,-y)(-1)^{x+y}$$

Since

$$(-1)^{x+y}(-1)^{x+y} = (-1)^{2x+2y} = 1$$

We obtain

$$h(x,y) = F(-x,-y)$$

or, for discrete images

$$h(x,y) = F[(-x) \bmod M, (-y) \bmod N]$$

The transformation $F(x,y) \rightarrow F(-x,-y)$ is a 180° rotation about the origin (equivalently, a point reflection through the origin). Therefore, final image = original image rotated by 180°

The D and P exchange positions and are rotated by 180° while the centrally symmetric I appears essentially unchanged.

Q.447. The Spatial filter averages the four closest neighbors of each pixel and excludes the center pixel. Given 3×3 kernel. The four neighbors are the pixels above, below, left and right of the center.

$$h(x,y) = \frac{1}{4}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

So,

$$g(x,y) = \frac{1}{4}\left[f(x-1,y)+f(x+1,y)+f(x,y-1)+f(x,y+1)\right]$$

a) For a 2D spatial filter, the frequency response is the 2-D fourier transform of the kernel.

$$H(U,V) = \sum_x \sum_y h(x,y)\, e^{-j2\pi(Ux+Vy)}$$

Taking the center of the kernel as (0,0) the four non-zero coefficients are at

(-1,0), (1,0), (0,-1), (0,1)

Therefore

$$H(U,V) = \frac{1}{4}\left[e^{-j2\pi U} + e^{j2\pi U} + e^{-j2\pi V} + e^{j2\pi V}\right.$$

using $e^{j\theta} + e^{-j\theta} = 2\cos\theta$.

We get

$$H(U,V) = \frac{1}{4}\left[2\cos(2\pi U) + 2\cos(2\pi V)\right]$$

$$H(U,V) = \frac{1}{2}\left[\cos(2\pi U) + \cos(2\pi V)\right]$$

This is the required frequency domain transfer function.

If angular frequencies W_1, W_2 are used, the equivalent expression is

$$H(W_1, W_2) = \frac{1}{2}[\cos W_1 + \cos W_2)$$

b) Consider the response at the zero frequency

$$\therefore H(0,0) = \frac{1}{2}[\cos 0 + \cos 0]$$

Since $\cos 0 = 1$

We obtain

$$H(0,0) = 1$$

Thus the DC component (zero-frequency component) passes completely.

Now consider frequencies close to the origin using the approximation

$$\cos \theta \approx 1 - \frac{\theta^2}{2}$$

We have,

$$H(U,V) \approx \frac{1}{2}\left[1 - \frac{(2\pi U)^2}{2} + 1 - \frac{(2\pi V)^2}{2}\right]$$

Therefore

$$H(U,V) \approx 1 - \pi^2(U^2 + V^2)$$

Hence, as U and V move away from zero,

$H(U,V)$ decreases from its maximum value of 1 near the origin. Low-frequency components are therefore passed with relatively high gain, while many higher-frequency components are attenuated. Thus, the filter behaves as a lowpass smoothing filter.

Q 4.57 A Gaussian lowpass filter smooth (a) the image by preserving low spatial frequencies and suppressing high spatial frequencies. It's transfer function is

$$H_{LP}(U,V) = e^{\frac{-D^2(U,V)}{2D_0^2}}$$

Where $D(U,V)$ is the distance from the frequency-domain origin and $D_0 = 25$.
Effect: Fine details and sharp edges are reduced while slowly varying intensity regions are preserved.

Step 2 Understand Gaussian highpass filtering
A Gaussian highpass filter removes low spatial frequencies.
Its transfer function is

$$H_{HP}(U,V) = 1 - e^{\frac{-D^2(U,V)}{2D_0^2}}$$

Effect: It highlights rapid intensity changes, such as edges of bones, fingers and the ring.

Step 3 Apply both filters in sequence. The highpass filter is applied to the already lowpass-filtered image.

The combined transfer function is $H(U,V) = H_{LP}(U,V)$

$$H_{HP}(U,V)$$

$$H(U,V) = e^{\frac{-D^2}{2D_0^2}}\left(1 - e^{\frac{-D^2}{2D_0^2}}\right)$$

This combined filter suppresses both very low and very high frequencies retaining mainly a band of intermediate frequencies.
Although a highpass filter removes the DC component and suppresses the low frequencies the interior of the ring does not nessesarily remain constant after lowpass filtering.
The metal ring has strong intensity differences between itself and the surrounding finger.
The Gaussian lowpass filter blurs the ring boundaries.
Since the ring is relatively narrow blurred effects of its two boundaries overlap.
The subsequent highpass filter detects the remaining intensitity variations, and the responses from the two boundaries overlap to produce a bright, nearly solid region in the displayed result.

b) No, the result would theoretically be the same if we applied the Gaussian filterpass filter first and Gaussian lowpass filter second.
For linear shift-invariant filters, convolution is commutative.

$$f * h_{LP} * h_{HP} = f * h_{HP} * h_{LP}$$

Similarly in the frequency domain.

$$G(U,V) = F(U,V)\, H_{LP}(U,V)\, H_{HP}(U,V)$$

Because multiplication is commutative.

$$H_{LP}(U,V)\, H_{HP}(U,V) = H_{HP}(U,V)\, H_{LP}(U,V)$$